Lesson 47: TD(0) and n-Step TD

AI/ML Mathematics

Contents

1 Motivation and Overview **2**

2 References **3**

3 Monte Carlo Prediction as Baseline **3**

4 TD(0) Prediction **4**
- 4.1 Properties of the TD Error 4
- 4.2 Convergence of TD(0) 5

5 Bias–Variance Analysis **6**

6 n-Step TD Prediction **7**
- 6.1 Telescoping Decomposition 8
- 6.2 Convergence of n-Step TD 8

7 Exercises **9**

8 Summary **10**

9 Further Reading **10**

1 Motivation and Overview

Consider an agent navigating a large gridworld with thousands of states, receiving a reward only upon reaching a distant goal. After each episode, the Monte Carlo approach updates every visited state using the realized return—but those returns are sums of dozens of random transitions, producing estimates with enormous variance. Worse, the agent must wait until the episode terminates before any learning occurs. In continuing (non-episodic) environments, Monte Carlo methods cannot be applied at all.

Temporal-difference (TD) learning resolves both problems by *bootstrapping*: updating value estimates using other value estimates rather than waiting for a complete return. The simplest variant, TD(0), adjusts the estimate of each state immediately after a single transition, using the observed reward and the current estimate of the next state as a proxy for the full future return. The n-step extension interpolates between the one-step bootstrap of TD(0) and the full-episode return of Monte Carlo, providing a tunable bias–variance tradeoff. The tools developed in this lesson will let us prove that these bootstrapped updates converge to the true value function despite the apparent circularity of updating estimates from estimates.

Where TD(0) and n-Step TD Appear

- **TD(0)** is the prediction engine inside SARSA and Q-learning (Lesson 49), where the TD error drives value updates.
- **n-step returns** underpin Generalized Advantage Estimation (GAE), the variance reduction technique used by PPO and TRPO for policy gradient estimation.
- **TD errors** form the building blocks of eligibility traces and TD(λ) (Lesson 48), connecting the forward and backward views of credit assignment.
- **Bias–variance analysis** of TD targets guides hyperparameter selection (n, λ) in every practical deep RL implementation.

Prerequisites. Markov decision processes: state-value function $V^\pi(s) = \mathbb{E}_\pi[\sum_{t=0}^\infty \gamma^t R_{t+1} \mid S_0 = s]$, Bellman expectation equation, Bellman operator $\mathcal{T}^\pi$ and its γ-contraction property in $\|\cdot\|_\infty$ (Lessons 45–46). Stochastic approximation: the Robbins–Monro theorem with martingale noise, step-size conditions $\sum \alpha_n = \infty$, $\sum \alpha_n^2 < \infty$ (Lesson 31). Conditional expectation and the law of total variance (Lesson 7).

Learning objectives.

- *State* the TD(0) update rule and explain how it differs from the Monte Carlo update through bootstrapping.
- *Prove* that the TD error δ_t^π under the true value function is a martingale difference sequence.
- *Derive* the convergence of TD(0) by casting it as a stochastic approximation of the Bellman equation.
- *Analyze* the bias–variance tradeoff between TD(0) and Monte Carlo targets.
- *Define* the n-step return and *prove* the telescoping decomposition into TD errors.
- *Prove* convergence of n-step TD for any fixed $n \geq 1$.

2 References

- Sutton and Barto [2018, Chapter 6] — Introduction to TD prediction; TD(0) algorithm and examples.
- Sutton and Barto [2018, Chapter 7] — n-step bootstrapping methods and the bias–variance spectrum.
- Bertsekas and Tsitsiklis [1996, Chapter 4] — Convergence of temporal difference methods via stochastic approximation.
- Szepesvári [2010, Section 4] — Compact treatment of TD convergence under the stochastic approximation framework.
- Tsitsiklis [1994] — Asynchronous stochastic approximation and the convergence machinery for TD and Q-learning.
- Sutton [1988] — The original paper formalizing TD learning and the TD(λ) family.

3 Monte Carlo Prediction as Baseline

We begin with Monte Carlo (MC) prediction to establish the baseline against which TD methods will be compared. Throughout this lesson, we work with a finite MDP $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$ with $\gamma \in [0, 1)$, a stationary policy π, and the return $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$.

Definition 3.1 (Monte Carlo Update)**.** Given an observed return G_t from state S_t, the *every-visit MC update* is

$$V(S_t) \leftarrow V(S_t) + \alpha\big[G_t - V(S_t)\big]. \tag{1}$$

Proposition 3.2 (Convergence of MC Prediction)**.** *Suppose every state $s \in \mathcal{S}$ is visited infinitely often and the step sizes satisfy $\sum_n \alpha_n(s) = \infty$, $\sum_n \alpha_n^2(s) < \infty$ for each s. Then the MC update (1) converges to $V^\pi(s)$ almost surely.*

Proof. Since $\mathbb{E}_\pi[G_t \mid S_t = s] = V^\pi(s)$, the MC update is a Robbins–Monro iteration with driving function $h(V(s)) = V^\pi(s) - V(s)$ and noise $w_t = G_t - V^\pi(s)$. The noise has zero conditional mean: $\mathbb{E}[w_t \mid S_t = s] = 0$. The mapping h has unique root $V(s) = V^\pi(s)$, and the step-size conditions are satisfied by assumption. By the Robbins–Monro theorem (Lesson 31), $V_n(s) \to V^\pi(s)$ almost surely. □

Lemma 3.3 (Variance of the MC Return)**.** *For deterministic rewards $r(s, a, s')$, the variance of the return satisfies*

$$\mathrm{Var}_\pi(G_t \mid S_t = s) = \gamma^2 \mathbb{E}_\pi\big[\mathrm{Var}_\pi(G_{t+1} \mid S_{t+1}) \,\big|\, S_t = s\big] + \mathrm{Var}_\pi\big(r(S_t, A_t, S_{t+1}) + \gamma V^\pi(S_{t+1}) \,\big|\, S_t = s\big). \tag{2}$$

Proof. Write $G_t = R_{t+1} + \gamma G_{t+1}$. By the law of total variance conditioned on (S_t, A_t, S_{t+1}):

$$\mathrm{Var}_\pi(G_t \mid S_t) = \mathbb{E}_\pi\big[\mathrm{Var}_\pi(G_t \mid S_t, A_t, S_{t+1}) \,\big|\, S_t\big] + \mathrm{Var}_\pi\big(\mathbb{E}_\pi[G_t \mid S_t, A_t, S_{t+1}] \,\big|\, S_t\big).$$

For deterministic rewards, the inner conditional variance reduces to $\gamma^2 \mathrm{Var}_\pi(G_{t+1} \mid S_{t+1})$ and the inner conditional expectation is $r(S_t, A_t, S_{t+1}) + \gamma V^\pi(S_{t+1})$, yielding (2). □

MC Variance Accumulates Over the Entire Trajectory

Equation (2) shows that $\mathrm{Var}_\pi(G_t \mid S_t)$ includes a γ^2-weighted contribution from $\mathrm{Var}_\pi(G_{t+1} \mid S_{t+1})$, which in turn depends on $\mathrm{Var}_\pi(G_{t+2} \mid S_{t+2})$, and so on. The variance accumulates from *all* future time steps, growing substantially when γ is close to 1. This motivates the bootstrapping approach of TD methods.

The recursive structure of (2) makes clear that the MC target G_t incorporates randomness from the entire trajectory. We now introduce TD(0), which eliminates all future variance beyond a single transition.

4 TD(0) Prediction

Definition 4.1 (TD Error)**.** Given a value estimate V, the *temporal-difference error* at time t is

$$\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t). \tag{3}$$

When $V = V^\pi$, we write $\delta_t^\pi = R_{t+1} + \gamma V^\pi(S_{t+1}) - V^\pi(S_t)$.

Definition 4.2 (TD(0) Update Rule)**.** The *TD(0) update* modifies the value estimate upon observing the transition $(S_t, A_t, R_{t+1}, S_{t+1})$:

$$V(S_t) \leftarrow V(S_t) + \alpha\,\delta_t = V(S_t) + \alpha\big[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\big]. \tag{4}$$

Remark 4.3 (Bootstrapping). The key distinction from MC is that TD(0) uses the *bootstrapped target* $R_{t+1} + \gamma V(S_{t+1})$ instead of the full return G_t. This replaces all future randomness (from G_{t+1}) with the current estimate $V(S_{t+1})$, reducing variance at the cost of introducing bias when $V \neq V^\pi$.

4.1 Properties of the TD Error

Lemma 4.4 (Unbiasedness of the TD Error)**.** *Under policy π, the TD error computed with the true value function has zero conditional expectation:*

$$\mathbb{E}_\pi\big[\delta_t^\pi \mid S_t = s\big] = 0 \quad \textit{for all } s \in \mathcal{S}. \tag{5}$$

That is, $\{\delta_t^\pi\}$ is a martingale difference sequence with respect to the filtration generated by $(S_0, A_0, R_1, S_1, \ldots)$.

Proof. Compute the conditional expectation directly:

$$\begin{aligned}
\mathbb{E}_\pi\big[\delta_t^\pi \mid S_t = s\big] &= \mathbb{E}_\pi\big[R_{t+1} + \gamma V^\pi(S_{t+1}) - V^\pi(S_t) \mid S_t = s\big] \\
&= \mathbb{E}_\pi\big[R_{t+1} + \gamma V^\pi(S_{t+1}) \mid S_t = s\big] - V^\pi(s) \\
&= \sum_a \pi(a \mid s) \sum_{s'} \mathcal{P}(s' \mid s, a)\big[r(s,a,s') + \gamma V^\pi(s')\big] - V^\pi(s) \\
&= (\mathcal{T}^\pi V^\pi)(s) - V^\pi(s) = V^\pi(s) - V^\pi(s) = 0,
\end{aligned} \tag{6}$$

where the last equality uses the fact that V^π is the fixed point of $\mathcal{T}^\pi$. □

Return as Telescoping Sum of TD Errors

Lemma 4.5 (Telescoping Decomposition)**.** *Under the true value function V^π, the predic-*

tion error decomposes as

$$G_t - V^\pi(S_t) = \sum_{k=0}^{\infty} \gamma^k \delta^\pi_{t+k}. \tag{7}$$

Proof. Expand the right-hand side:

$$\begin{aligned}\sum_{k=0}^{\infty} \gamma^k \delta^\pi_{t+k} &= \sum_{k=0}^{\infty} \gamma^k \left[R_{t+k+1} + \gamma V^\pi(S_{t+k+1}) - V^\pi(S_{t+k})\right] \\ &= \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} + \sum_{k=0}^{\infty} \gamma^{k+1} V^\pi(S_{t+k+1}) - \sum_{k=0}^{\infty} \gamma^k V^\pi(S_{t+k}). \end{aligned} \tag{8}$$

Re-index the second sum with $j = k + 1$:

$$\begin{aligned} &= G_t + \sum_{j=1}^{\infty} \gamma^j V^\pi(S_{t+j}) - \sum_{k=0}^{\infty} \gamma^k V^\pi(S_{t+k}) \\ &= G_t + \sum_{k=1}^{\infty} \gamma^k V^\pi(S_{t+k}) - V^\pi(S_t) - \sum_{k=1}^{\infty} \gamma^k V^\pi(S_{t+k}) \\ &= G_t - V^\pi(S_t), \end{aligned} \tag{9}$$

where the telescoping cancellation eliminates all intermediate value terms. The boundary term $\lim_{k\to\infty} \gamma^k V^\pi(S_{t+k}) = 0$ since $\gamma < 1$ and V^π is bounded on the finite state space. □

TD Errors as Prediction Errors

Lemma 4.5 shows that the return minus the value — the total prediction error — equals the sum of discounted one-step TD errors. This decomposition is the mathematical foundation for:

- The forward view of TD(λ) (Lesson 48), where exponentially weighted sums of TD errors define the λ-return.
- Generalized Advantage Estimation (GAE), which uses truncated versions of this sum for low-variance policy gradient estimation.

4.2 Convergence of TD(0)

We now prove the central result of this lesson: TD(0) converges to the true value function almost surely.

Assumption 4.6 (TD(0) Convergence Conditions)**.** We assume:

1. The MDP has finite state and action spaces.
2. The policy π is stationary.
3. Every state $s \in \mathcal{S}$ is visited infinitely often.
4. The step sizes satisfy $\sum_{n=0}^{\infty} \alpha_n(s) = \infty$ and $\sum_{n=0}^{\infty} \alpha_n^2(s) < \infty$ for each $s \in \mathcal{S}$.

Convergence of TD(0)

Theorem 4.7 (Convergence of TD(0))**.** *Under Assumption 4.6, the TD(0) update (4) converges to V^π almost surely:*

$$V_n(s) \xrightarrow{a.s.} V^\pi(s) \quad \textit{for all } s \in \mathcal{S}. \tag{10}$$

Proof. We cast TD(0) as a stochastic approximation of the Bellman equation and verify all conditions of the Robbins–Monro theorem (Lesson 31, Theorem 3.1). The proof follows Tsitsiklis [1994] and Jaakkola et al. [1994].

Step 1: Express as stochastic approximation. Fix a state s and let $n_k(s)$ denote the time of the k-th visit to s. At these visits, the TD(0) update reads

$$V_{k+1}(s) = V_k(s) + \alpha_k(s)\big[R_{n_k+1} + \gamma V_k(S_{n_k+1}) - V_k(s)\big]. \tag{11}$$

Decompose the update target into its expected value plus noise:

$$R_{n_k+1} + \gamma V_k(S_{n_k+1}) = (\mathcal{T}^\pi V_k)(s) + w_k(s), \tag{12}$$

where $w_k(s) = R_{n_k+1} + \gamma V_k(S_{n_k+1}) - (\mathcal{T}^\pi V_k)(s)$.

Step 2: Verify martingale noise. Let $\mathcal{F}_k$ denote the filtration containing all information up to the k-th visit to s (including V_k and $S_{n_k} = s$). Then:

$$\begin{aligned}
\mathbb{E}[w_k(s) \mid \mathcal{F}_k] &= \mathbb{E}\big[R_{n_k+1} + \gamma V_k(S_{n_k+1}) \mid S_{n_k} = s, V_k\big] - (\mathcal{T}^\pi V_k)(s) \\
&= \sum_a \pi(a \mid s) \sum_{s'} \mathcal{P}(s' \mid s, a)\big[r(s,a,s') + \gamma V_k(s')\big] - (\mathcal{T}^\pi V_k)(s) \\
&= (\mathcal{T}^\pi V_k)(s) - (\mathcal{T}^\pi V_k)(s) = 0.
\end{aligned} \tag{13}$$

Thus $\{w_k(s)\}$ is a martingale difference sequence with respect to $\{\mathcal{F}_k\}$.

Step 3: Identify the expected update direction. Substituting (12) into (11):

$$V_{k+1}(s) = V_k(s) + \alpha_k(s)\big[(\mathcal{T}^\pi V_k)(s) - V_k(s) + w_k(s)\big]. \tag{14}$$

The deterministic driving function is $h(V)(s) = (\mathcal{T}^\pi V)(s) - V(s)$, which has the unique root $V = V^\pi$ (the fixed point of $\mathcal{T}^\pi$).

Step 4: Verify stability via contraction. Since $\mathcal{T}^\pi$ is a γ-contraction in $\|\cdot\|_\infty$ (Lessons 45–46), for any $\alpha \in (0, 1]$:

$$\begin{aligned}
\|V + \alpha\, h(V) - V^\pi\|_\infty &= \|(1-\alpha)V + \alpha\, \mathcal{T}^\pi V - V^\pi\|_\infty \\
&= \|(1-\alpha)(V - V^\pi) + \alpha(\mathcal{T}^\pi V - \mathcal{T}^\pi V^\pi)\|_\infty \\
&\le (1-\alpha)\|V - V^\pi\|_\infty + \alpha\,\gamma\,\|V - V^\pi\|_\infty \\
&= \big(1 - \alpha(1-\gamma)\big)\|V - V^\pi\|_\infty.
\end{aligned} \tag{15}$$

Since $\alpha(1-\gamma) > 0$, the expected update contracts the error toward zero, confirming that h drives iterates toward V^π.

Step 5: Bound the noise variance. By finiteness of the MDP, there exists $R_{\max} < \infty$ such that $|R_{t+1}| \le R_{\max}$. The value iterates remain bounded throughout: $\|V_k\|_\infty \le R_{\max}/(1-\gamma)$, which can be established by induction using (15). Therefore:

$$\mathbb{E}\big[\|w_k(s)\|^2 \mid \mathcal{F}_k\big] \le C\big(1 + \|V_k\|_\infty^2\big) \tag{16}$$

for a constant C depending on $R_{\max}$, γ, and $|\mathcal{S}|$.

All five conditions of the Robbins–Monro theorem are verified: (1) step sizes, (2) martingale noise, (3) unique root of h, (4) stability via contraction, (5) bounded noise variance. Therefore $V_n(s) \to V^\pi(s)$ almost surely for all $s \in \mathcal{S}$. □

Remark 4.8 (Convergence Despite Bootstrapping Bias). At any finite time, $V(S_{t+1}) \neq V^\pi(S_{t+1})$, so the TD target $R_{t+1} + \gamma V(S_{t+1})$ is a biased estimate of $V^\pi(S_t)$. Why does TD(0) still converge? The contraction property (Step 4) provides the answer: the expected update direction $\mathcal{T}^\pi V - V$ always points toward V^π, and the contraction ensures that the bootstrapping loop is stable — the bias shrinks as V approaches V^π, creating a virtuous cycle.

Example 4.9 (Two-State MDP)**.** Consider an MDP with $\mathcal{S} = \{A, B\}$, a single action, deterministic transitions $A \to B \to A \to \cdots$, rewards $r(A) = 1$, $r(B) = 0$, and $\gamma = 0.9$. The true values are $V^\pi(A) = 100/19 \approx 5.26$ and $V^\pi(B) \approx 4.74$. Starting from $V_0(A) = V_0(B) = 0$ with $\alpha = 0.1$, the first TD(0) update at A gives $V_1(A) = 0.1(1 + 0) = 0.1$, gradually refining toward the true values through bootstrapping.

5 Bias–Variance Analysis

We now make precise the tradeoff between bias and variance that distinguishes TD(0) from MC.

Proposition 5.1 (Bias of the TD(0) Target)**.** *The conditional expectation of the TD(0) target is*

$$\mathbb{E}_\pi\big[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s\big] = (\mathcal{T}^\pi V)(s), \tag{17}$$

which equals $V^\pi(s)$ if and only if $V = V^\pi$. Thus the TD(0) target is biased *whenever $V \neq V^\pi$.*

Proof. Direct computation:

$$\mathbb{E}_\pi\big[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s\big] = \sum_a \pi(a \mid s) \sum_{s'} \mathcal{P}(s' \mid s, a)\big[r(s, a, s') + \gamma V(s')\big] = (\mathcal{T}^\pi V)(s). \tag{18}$$

Since V^π is the unique fixed point of $\mathcal{T}^\pi$, $(\mathcal{T}^\pi V)(s) = V^\pi(s)$ for all s if and only if $V = V^\pi$. □

Proposition 5.2 (Variance Reduction by TD(0))**.** *For deterministic rewards, the variance of the TD(0) target satisfies*

$$\mathrm{Var}_\pi\big(R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s\big) = \mathrm{Var}_\pi\big(r(s, A_t, S_{t+1}) + \gamma V(S_{t+1}) \mid S_t = s\big), \tag{19}$$

which depends only on the randomness of the next transition (A_t, S_{t+1}). In contrast, $\mathrm{Var}_\pi(G_t \mid S_t = s)$ depends on all future transitions.

Proof. Since $R_{t+1} + \gamma V(S_{t+1})$ is fully determined by (S_t, A_t, S_{t+1}), its variance involves only the randomness of the next action and transition. Using the law of total variance with $V = V^\pi$:

$$\mathrm{Var}_\pi(G_t \mid S_t) = \underbrace{\mathrm{Var}_\pi\big(R_{t+1} + \gamma V^\pi(S_{t+1}) \mid S_t\big)}_{\text{one-step variance}} + \gamma^2 \underbrace{\mathbb{E}_\pi\big[\mathrm{Var}_\pi(G_{t+1} \mid S_{t+1}) \mid S_t\big]}_{\text{future variance} \geq 0}. \tag{20}$$

The TD(0) target captures only the first term, eliminating the non-negative future variance entirely. □

Bias–Variance Tradeoff: MC vs TD(0)

	Bias	**Variance**
MC (G_t)	Zero	High (all future steps)
TD(0) ($R_{t+1} + \gamma V(S_{t+1})$)	Nonzero (when $V \neq V^\pi$)	Low (one step)

The n-step TD methods developed next provide a continuum between these two extremes.

Example 5.3 (Variance Comparison)**.** In a chain with $\gamma = 0.99$ (effective horizon ≈ 100) and per-step reward variance σ^2, the MC return variance is approximately $\sum_{k=0}^{99} \gamma^{2k}\sigma^2 \approx 50\sigma^2$, while the TD(0) target variance is approximately σ^2 — a 50-fold reduction.

6 n-Step TD Prediction

The n-step TD method interpolates between TD(0) (one-step bootstrap) and MC (full return) by using n actual rewards before bootstrapping.

Definition 6.1 (n-Step Return)**.** The *n-step return* from time t is

$$G_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n V(S_{t+n}). \tag{21}$$

For $n = 1$, this reduces to the TD(0) target $R_{t+1} + \gamma V(S_{t+1})$. In the limit $n \to \infty$ (with episode length T), the bootstrap term $\gamma^n V(S_{t+n}) \to 0$ and $G_t^{(n)} \to G_t$.

Definition 6.2 (n-Step TD Update)**.** The *n-step TD update* is

$$V(S_t) \leftarrow V(S_t) + \alpha\big[G_t^{(n)} - V(S_t)\big]. \tag{22}$$

Example 6.3 (n-Step Returns in a Chain)**.** Consider a chain $A \to B \to C \to D$ with $\gamma = 0.9$, rewards $r(A) = 1, r(B) = 0, r(C) = 2$, and estimates $V(B) = 3, V(C) = 4, V(D) = 2$. From state A: $G_A^{(1)} = 1 + 0.9(3) = 3.7$, $G_A^{(2)} = 1 + 0 + 0.81(4) = 4.24$, $G_A^{(3)} = 1 + 0 + 1.62 + 0.729(2) = 4.078$. Each uses more actual rewards and less bootstrapping.

6.1 Telescoping Decomposition

Lemma 6.4 (n-Step Return as Telescoping Sum of TD Errors)**.** *The n-step prediction error decomposes as*

$$G_t^{(n)} - V(S_t) = \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}, \tag{23}$$

where $\delta_{t+k} = R_{t+k+1} + \gamma V(S_{t+k+1}) - V(S_{t+k})$.

Proof. Expand the right-hand side:

$$\begin{aligned}\sum_{k=0}^{n-1} \gamma^k \delta_{t+k} &= \sum_{k=0}^{n-1} \gamma^k \big[R_{t+k+1} + \gamma V(S_{t+k+1}) - V(S_{t+k})\big] \\ &= \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \sum_{k=0}^{n-1} \gamma^{k+1} V(S_{t+k+1}) - \sum_{k=0}^{n-1} \gamma^k V(S_{t+k}).\end{aligned} \tag{24}$$

Re-index the second sum with $j = k + 1$:

$$\begin{aligned}&= \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \sum_{j=1}^{n} \gamma^j V(S_{t+j}) - \sum_{k=0}^{n-1} \gamma^k V(S_{t+k}) \\ &= \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n V(S_{t+n}) - V(S_t) = G_t^{(n)} - V(S_t),\end{aligned} \tag{25}$$

where the telescoping cancellation eliminates all intermediate value terms $\gamma^k V(S_{t+k})$ for $k = 1, \ldots, n-1$. □

Remark 6.5 (Connecting to the Infinite Telescoping). Lemma 6.4 is the finite-n analogue of Lemma 4.5. As $n \to \infty$, the n-step decomposition (23) approaches the infinite decomposition (7), with the bootstrap term $\gamma^n V(S_{t+n}) \to 0$ absorbed into the tail of the TD error sum.

6.2 Convergence of n-Step TD

Theorem 6.6 (Convergence of n-Step TD)**.** *Under Assumption 4.6, the n-step TD update (22) converges to V^π almost surely for any fixed $n \geq 1$.*

Proof. Define the n-step Bellman operator $\mathcal{T}_n^\pi : \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ by

$$(\mathcal{T}_n^\pi V)(s) = \mathbb{E}_\pi\big[G_t^{(n)} \mid S_t = s\big] = \mathbb{E}_\pi\left[\sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n V(S_{t+n}) \,\middle|\, S_t = s\right]. \tag{26}$$

Step 1: $\mathcal{T}_n^\pi$ is a γ^n-contraction. The operator $\mathcal{T}_n^\pi$ is the n-fold composition $(\mathcal{T}^\pi)^n$. Since $\mathcal{T}^\pi$ is a γ-contraction in $\|\cdot\|_\infty$, we have for any V_1, V_2:

$$\|\mathcal{T}_n^\pi V_1 - \mathcal{T}_n^\pi V_2\|_\infty = \|(\mathcal{T}^\pi)^n V_1 - (\mathcal{T}^\pi)^n V_2\|_\infty \leq \gamma^n \|V_1 - V_2\|_\infty. \tag{27}$$

The unique fixed point of $\mathcal{T}_n^\pi$ is V^π (since $\mathcal{T}^\pi V^\pi = V^\pi$ implies $(\mathcal{T}^\pi)^n V^\pi = V^\pi$).

Step 2: Cast as stochastic approximation. The n-step TD update at the k-th visit to state s has the form

$$V_{k+1}(s) = V_k(s) + \alpha_k(s)\big[(\mathcal{T}_n^\pi V_k)(s) - V_k(s) + w_k^{(n)}(s)\big], \tag{28}$$

where $w_k^{(n)}(s) = G_t^{(n)} - (\mathcal{T}_n^\pi V_k)(s)$ is zero-mean noise (by the same argument as Step 2 in the TD(0) proof).

Step 3: Verify remaining conditions. The driving function $h_n(V)(s) = (\mathcal{T}_n^\pi V)(s) - V(s)$ has unique root V^π. The γ^n-contraction ensures stability (indeed, the contraction rate is *stronger* than TD(0)). The noise variance is bounded since rewards are bounded and n is finite. Step-size conditions hold by assumption.

All conditions of the Robbins–Monro theorem are satisfied, so $V_k(s) \to V^\pi(s)$ almost surely for all s. □

The Spectrum from TD(0) to MC

The n-step TD family provides a continuum parameterized by n:

- $n = 1$: TD(0) — maximal bias, minimal variance, fastest per-step computation.
- $n = T$ (episode length): MC — zero bias, maximal variance, requires complete episodes.
- Intermediate n: trades off bias and variance; optimal n depends on the problem.

In deep RL practice, moderate values of n (typically 5–20) combined with value function baselines yield the best empirical performance.

Example 6.7 (Contraction Rate vs. Variance)**.** For $\gamma = 0.99$: TD(0) contracts at rate 0.99, while 10-step TD contracts at rate $0.99^{10} \approx 0.904$ (faster). But the 10-step target has higher variance from 10 random transitions. The optimal n balances faster contraction against noisier updates.

7 Exercises

Exercise 7.1 (TD Error Variance)**.** (∗)
Consider a finite MDP with deterministic rewards. Show that the variance of the TD error under the true value function satisfies

$$\mathrm{Var}_\pi(\delta_t^\pi \mid S_t = s) = \mathrm{Var}_\pi\big(R_{t+1} + \gamma V^\pi(S_{t+1}) \mid S_t = s\big).$$

Interpret this as the one-step variance that TD(0) uses.

Exercise 7.2 (MC as n-Step TD). $(*)$
For an episodic MDP with episode length T starting from time $t = 0$, verify directly from Definition 6.1 that $G_0^{(T)} = G_t[0]$ when $V(S_T) = 0$ (terminal state has zero value).

Exercise 7.3 (Contraction Rate of the n-Step Operator). $(**)$
Let $\gamma = 0.95$ and $n = 5$. Compute the contraction rate γ^n and determine how many applications of $\mathcal{T}_n^\pi$ are needed to reduce the initial error $\|V_0 - V^\pi\|_\infty$ by a factor of 10^{-3}.
Hint: Solve $\gamma^{nk} \leq 10^{-3}$ for k.

Exercise 7.4 (Stochastic Approximation Decomposition for $n = 2$). $(**)$
Write out the full stochastic approximation decomposition for the 2-step TD update: express $G_t^{(2)} - V(S_t)$ as $(\mathcal{T}_2^\pi V)(S_t) - V(S_t) + w_t^{(2)}$, and verify explicitly that $\mathbb{E}[w_t^{(2)} \mid S_t, V] = 0$.

Exercise 7.5 (Bias Quantification). $(***)$
Let V be an arbitrary value function estimate with $\|V - V^\pi\|_\infty \leq \varepsilon$. Show that the bias of the n-step return satisfies

$$\left|\mathbb{E}_\pi\left[G_t^{(n)} \mid S_t = s\right] - V^\pi(s)\right| \leq \gamma^n \varepsilon \quad \text{for all } s \in \mathcal{S}.$$

Conclude that larger n reduces the bias geometrically.
Hint: Use the fact that $\mathcal{T}_n^\pi$ is a γ^n-contraction with fixed point V^π.

Exercise 7.6 (Variance of the n-Step Return). $(***)$
Assuming $V = V^\pi$ and deterministic rewards, derive a recursive formula for $\mathrm{Var}_\pi(G_t^{(n)} \mid S_t = s)$ in terms of one-step transition variances. Show that $\mathrm{Var}_\pi(G_t^{(n)} \mid S_t)$ is non-decreasing in n, confirming that longer lookback increases variance.
Hint: Use the telescoping decomposition $G_t^{(n)} - V^\pi(S_t) = \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}^\pi$ and the fact that the δ_{t+k}^π are martingale differences.

8 Summary

- The **TD error** $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$ measures the one-step prediction discrepancy; under the true value function, it forms a *martingale difference sequence* ($\mathbb{E}[\delta_t^\pi \mid S_t] = 0$).
- The **telescoping decomposition** $G_t - V^\pi(S_t) = \sum_{k=0}^{\infty} \gamma^k \delta_{t+k}^\pi$ decomposes the total prediction error into a sum of discounted one-step TD errors — the foundation for TD(λ) and GAE.
- **TD(0)** converges to V^π almost surely under standard conditions, proved by casting it as a stochastic approximation of the Bellman equation: the Bellman operator's contraction property provides stability, and the martingale noise has bounded variance.
- **Bias–variance tradeoff**: MC targets are unbiased but have high variance (accumulated over all future steps); TD(0) targets have low variance (one step) but are biased when $V \neq V^\pi$.
- The **n-step return** $G_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n V(S_{t+n})$ interpolates between TD(0) ($n = 1$) and MC ($n \to \infty$), with contraction rate γ^n.
- **n-step TD converges** for any fixed $n \geq 1$ via the same stochastic approximation framework, with the n-step Bellman operator $\mathcal{T}_n^\pi = (\mathcal{T}^\pi)^n$ providing a γ^n-contraction.

9 Further Reading

- **Lesson 48** extends the n-step framework to TD(λ), which takes a geometrically weighted average of all n-step returns, and develops the theory of eligibility traces for efficient incremental computation.
- **Lesson 49** applies TD errors to the control setting, developing SARSA, Q-learning, and Expected SARSA with convergence proofs.
- Sutton [1988] is the original paper introducing TD learning and proving convergence for the linear case. The stochastic approximation approach used here follows Tsitsiklis [1994] and Jaakkola et al. [1994].
- Sutton and Barto [2018, Chapter 7] provides extensive empirical comparisons of n-step methods on benchmark tasks, illustrating the bias–variance tradeoff in practice.
- Borkar [2008] gives the definitive treatment of stochastic approximation via the ODE method, providing the theoretical foundations underlying all convergence proofs in this lesson.
- For the extension to function approximation, where TD methods can diverge (the "deadly triad"), see Tsitsiklis and Van Roy [1997] and Baird [1995].

References

Leemon Baird. Residual algorithms: Reinforcement learning with function approximation. In *Proceedings of the 12th International Conference on Machine Learning*, pages 30–37, 1995.

Dimitri P. Bertsekas and John N. Tsitsiklis. *Neuro-Dynamic Programming*. Athena Scientific, 1996.

Vivek S. Borkar. *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press, 2008.

Tommi Jaakkola, Michael I. Jordan, and Satinder P. Singh. On the convergence of stochastic iterative dynamic programming algorithms. *Neural Computation*, 6(6):1185–1201, 1994.

Richard S. Sutton. Learning to predict by the methods of temporal differences. *Machine Learning*, 3(1):9–44, 1988.

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.

Csaba Szepesvári. *Algorithms for Reinforcement Learning*. Morgan & Claypool, 2010.

John N. Tsitsiklis. Asynchronous stochastic approximation and Q-learning. *Machine Learning*, 16(3):185–202, 1994.

John N. Tsitsiklis and Benjamin Van Roy. An analysis of temporal-difference learning with function approximation. *IEEE Transactions on Automatic Control*, 42(5):674–690, 1997.